

CSMMetal202500001 Aegis Embark_ Vibration Mitigation Product

V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | **Date:** June 2026 | **Applies to document:** CSMMetal202500001 Aegis Embark_ Vibration Mitigation Product V2.0

Revision Necessity — Three Drivers

1. **Empirical data superseded** — 2024-2025 peer-reviewed publications invalidate V1.0 material property values
2. **Heliophysical risk recalibrated** — Solar Cycle 25 SSN ~137 (observed) vs ~115 (forecast used in V1.0); design basis updated accordingly
3. **Standards/regulations updated** — One or more referenced codes superseded; V1.0 compliance citations rendered obsolete

V1.0 vs V2.0 — Specific Parameter Changes

Parameter	V1.0	V2.0	Reason	Primary Source
Solar Cycle 25 risk framing	V1.0: Risk basis SSN ~115 (original SC25 forecast)	V2.0: Revised to SSN ~137 (NOAA SWPC Q1 2026 actuals); 10-year Carrington-class probability +20%	Observed solar activity exceeded forecast; design basis was non-conservative	NOAA SWPC Solar Cycle 25 Progression Report Q1 2026
MR fluid specification	V1.0: LORD MRF-132DG; yield stress 73 kPa	V2.0: LORD MRF-140CG; yield stress 80 kPa	9.6% yield stress improvement; higher stiffness modulation ratio	LORD Corp MRF-140CG Product Bulletin 2025
Piezoelectric material	V1.0: PZT-5H; d33=289 pC/N; Tc=193°C; lead-containing	V2.0: KNbO3-BaTiO3; d33=285 pC/N; Tc=310°C; lead-free	RoHS/REACH compliance — PZT restricted; Tc +117°C prevents thermal depolarization	TDK 2025 lead-free catalog; J. Am. Ceram. Soc. 108 (2025)

Parameter	V1.0	V2.0	Reason	Primary Source
Vibration exposure standards	V1.0: ISO 2631-1:1997; ISO 5349-1:2001	V2.0: ISO 2631-1:2025 (caution 0.5 m/s ² ; action 0.8 m/s ²); ISO 5349-1:2023	Both standards superseded; updated limits redefine compliance thresholds	ISO 2631-1:2025; ISO 5349-1:2023; WHO 2025
Composite matrix — recyclability	V1.0: Thermoset epoxy; irreversible crosslink; landfill end-of-life	V2.0: Elixir thermoplastic acrylic; thermal solvolysis 350°C; 95% fiber + 100% monomer recovery	Thermoset blocks Phoenix Protocol circular economy mandate	Arkema Elixir TDS 2025; Composites Part A 2024
EMI/FSS shielding layer	V1.0: AgNW/graphene screen-printed FSS; SE ~45 dB at 1 GHz	V2.0: Ti3C2Tx MXene spray (50 µm, 3x3 cm island pattern); SE 92 dB at 1 GHz	2.0x SE improvement; spray replaces photomask process; DC-isolated tile prevents GIC loop	ACS Appl. Mater. Interfaces 17/8 (2025) 12234

Impact If V1.0 Retained (Criticality Matrix)

Parameter Changed	Consequence of Non-Upgrade	Severity
Solar Cycle 25 risk framing	Probability underestimated; design basis not conservative for observed SC25 activity	MEDIUM
MR fluid specification	9.6% shielding deficit in active damping performance at design operating point	MEDIUM
Piezoelectric material	Lead restricted under RoHS Annex II; Tc 117°C lower = premature depolarization risk	HIGH

Parameter Changed	Consequence of Non-Upgrade	Severity
Vibration exposure standards	Compliance claims on obsolete standards; regulatory liability in product disputes	HIGH
Composite matrix — recyclability	No end-of-life recovery path; contradicts Phoenix Protocol; Phoenix Protocol breach	HIGH
EMI/FSS shielding layer	47 dB shielding deficit; AgNW susceptible to oxidation degradation over time	HIGH

Unchanged Elements

All design philosophy, fundamental equations (GIC induction, Joule heating, eddy current models), material selection rationale, product architecture, assembly methodology, and Phoenix Protocol structure are **carried forward unchanged** from V1.0. Only the specific parameters listed above were revised.

End of Versioning Delta | CSMMetal202500001 Aegis Embark_ Vibration Mitigation Product V1.0 archived; V2.0 is the active specification as of June 2026.